

Ana Yupanqui Ponce De Leon

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PROFILE

- Bachelor of Engineering (Honours) - Software student at UNSW, with a strong interest in Human-Computer Interaction, immersive systems, and AI-driven interactive design.
- Experienced in developing intelligent interactive systems using Unreal Engine, Unity, and machine learning frameworks.
- Interested in modelling user behaviour, emotion, and perception in real-time systems, particularly within VR and socially interactive environments.
- Proven ability to collaborate in multidisciplinary teams, manage complex tasks, and deliver innovative solutions under tight deadlines.
- Deep understanding of the mining industry, its challenges, and its importance to local and global economies. Passionate about leveraging technology to improve operational efficiency and safety in the mining and resources industry.

EDUCATION

Bachelor of Engineering (Honours) – Software The University of New South Wales (UNSW) , Sydney Honours Thesis (Supervised by Dr Benjamin Tag): • <i>Applying Latent Profile Analysis for User Clustering in Human-Computer Interaction.</i> • Investigating model-based clustering approaches to generate behaviourally meaningful user groupings for persona development and adaptive system design.	2022–ongoing.
Bachelor of Engineering (Honours) - Engineering Pathway The University of Adelaide , Adelaide • Transferred to UNSW after Year 1. • GPA 5.00 (Credit average for Engineering subjects)	2021–2022
Bachelor of Nursing (Year 1 completed) The University of Adelaide , Adelaide • GPA 6.00 (Distinction average).	2020–2021
SACE Certificate Sacred Heart College , Somerton Park	2017–2019

TECHNICAL EXPERIENCE - Software Projects

Interactive Survival Game – Unreal Engine 5 (AI, HCI, Immersive Systems)

Designed and developed a fully playable third-person survival game exploring human–companion interaction and noise-driven AI perception systems.

- Implemented multi-agent AI using Behaviour Trees and Unreal AI Perception (sight + hearing) to model reactive zombie agents driven by environmental stimuli.
- Designed a real-time stress modelling system for an AI companion (dog) based on proximity to threats using dynamic float scaling and state transitions.
- Developed command-based interaction mechanics (follow, stay, quiet) with stress-dependent duration scaling using interpolation functions.
- Integrated physics-based object interactions (projectile cans triggering AI hearing events via ReportNoiseEvent).
- Built dynamic UI elements (goal states, stress bar, command timer) driven by real-time system state.
- Applied immersive design principles including sound spatialisation, perception prioritisation (hearing > vision), and environmental storytelling.
- Packaged and deployed a Windows build for distribution.

Deep Learning Sea Turtle Image Segmentation Project- Python, PyTorch, OpenCV

Developed and evaluated deep learning pipelines for semantic and instance segmentation across 8,700+ images in a group setting.

- Implemented and compared **kNN, DeepLabV3+, Mask R-CNN, and YOLOv8-Seg** architectures.
- Achieved state-of-the-art performance (mAP@50 = 0.97; IoU = 0.972).
- Addressed class imbalance and signal noise using augmentation, preprocessing, and hyperparameter tuning.
- Conducted structured experimental comparisons across architectures.
- Optimised performance under hardware constraints.

“Roots and Shoots Foundation” – Figma Full Prototype

- Designed and prototyped a nonprofit charity website in **Figma**, collaborating in a 4-person team to apply the **user-centred design process**.
- Conducted **user research** (interviews, questionnaires, personas, requirements analysis) to inform functional and non-functional requirements.
- Implemented accessibility features (WCAG guidelines) and ran **formal usability tests**, iterating on the design to improve clarity and inclusivity for diverse users.

“GreenGotchi” – Sustainability-Focused Mobile App – Figma, Typescript, React Native, Python APIs.

- Collaborated in a cross-functional Agile team through three sprints to design and implement a **gamified mobile app** encouraging sustainable travel.
- Developed and integrated **five APIs** using NSW Transport data, enabling users to earn points for eco-friendly travel choices and care for virtual “tree pets.”
- Contributed to **front-end design in React Native** and **backend microservices**, overcoming challenges with API integration, environment setup, and CI/CD pipelines.
- Delivered a functional MVP, pitched to stakeholders in a **Shark Tank-style presentation**, and refined the product using usability testing and feedback.

Satellite Communication Simulation – Java

- Designed and implemented a medium-scale simulation of satellite and device communication around Jupiter using **Object-Oriented Programming** principles.
- Developed robust code with **programming by contract**, ensuring correctness through testing and iterative refactoring.

Dungeonmania Game Backend Refactoring – Java

- Inherited and refactored a large monolithic codebase for a dungeon-style game, improving design quality by resolving code smells and applying design patterns.
- Extended functionality to support new requirements, implementing iterative design and evolution of features.
- Collaborated with a partner to navigate an unfamiliar system, demonstrating adaptability and effective teamwork.

TECHNICAL SKILLS

- **Programming & ML:** Python (PyTorch, OpenCV), Java, JavaScript, React, SQL, C.
- **Immersive & Interactive Systems:** Unreal Engine 5, Unity, Behavioural Trees, AI Perception
- **Design & Tools:** Figma, Canva, Adobe XD, Microsoft Office Suite, Git., Confluence, Jira
- **Development Practices & Research:** Agile methodologies, version control, debugging, usability testing, and user research.
- **Certifications:** Basic Life Support, Safe Use of PPE, Infection Prevention and Control.

RECENT WORK EXPERIENCE

Phlebotomist – Clinpath Pathology (part of Sonic Healthcare Limited)

2025 – ongoing

- Performed precise clinical procedures while prioritizing and maintaining patient safety and comfort.
- Adherent to strict clinical protocols and maintained accurate patient records.
- Worked flexibly across multiple locations, adapting to varying operational demands.
- Built strong communication and teamwork skills with high attention to detail in high-pressure environments.

ADDITIONAL INTERESTS

- Dog training at the German Shepherd Club, SA.
- **Languages:** Spanish & English (native), French (conversational).